Paper 3 (75 marks)

1a	unit of pressure = (kg m s^{-2}) / m^2 = kg m^{-1} s^{-2} Unit of pressure is expressed in terms of the product and quotient of base units kg, m and s.	B1 B1
1b	$\frac{\Delta Q}{Q} = 4\frac{\Delta r}{r} + \frac{\Delta p_1 + \Delta p_2}{p_1 - p_2} + \frac{\Delta L}{L}$ $\frac{\Delta Q}{Q} = 4\frac{(0.04)}{(2.07)} + \frac{2+1}{(127-105)} + \frac{8}{200}$ $= 0.25$	C1 C1 A1
2a	Product of the force and the perpendicular distance from the point.	A1